Multi Agent Hospitality System

Course Name: Project-||

Institution Name: Medicaps University – Datagami Skill Based Course

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1. Problem Statement & Objectives

1.1. Problem Statement

The current landscape of digital travel planning is plagued by "Information Overload" and "Static Recommendations." Travelers typically spend between 3 to 5 hours toggling between multiple platforms—hotel booking sites, blogs, map services, and review portals—to piece together a single itinerary. Traditional AI chatbots often exacerbate this by providing hallucinated data, recommending permanently closed venues, or ignoring specific duration and luxury constraints. There is a critical lack of a unified, "agentic" system that can autonomously research, validate, and synthesize real-time web data into a coherent, professional itinerary that respects the nuances of high-end hospitality.

1.2. Project Objectives

- **Architect a Multi-Agent System:** Develop a modular graph-based architecture where specialized agents (Gatekeeper, Researcher, Writer) collaborate to solve complex planning tasks.
- **Ensure Real-Time Accuracy:** Integrate live web-search capabilities to ensure itineraries are based on current availability, events, and seasonal trends rather than outdated training data.
- **Implement Domain Guardrailing:** Create a robust filtering mechanism that prevents the AI from engaging in non-travel related topics, maintaining its "Concierge" persona.
- **Optimize Performance:** Leverage high-speed inference engines to deliver multi-step agentic responses in under 5 seconds, ensuring a premium user experience.

1.3. Scope of the Project

The project scope encompasses the backend agentic orchestration using LangGraph, real-time data retrieval via the Tavily Search API, and a polished Full-Stack web interface using Django. It focuses on generating 1-to-7 day itineraries for global destinations with a specific

emphasis on luxury and "bespoke" experiences. It excludes direct booking/payment processing but provides the necessary structured data for a traveler to proceed with bookings.

2. Proposed Solution

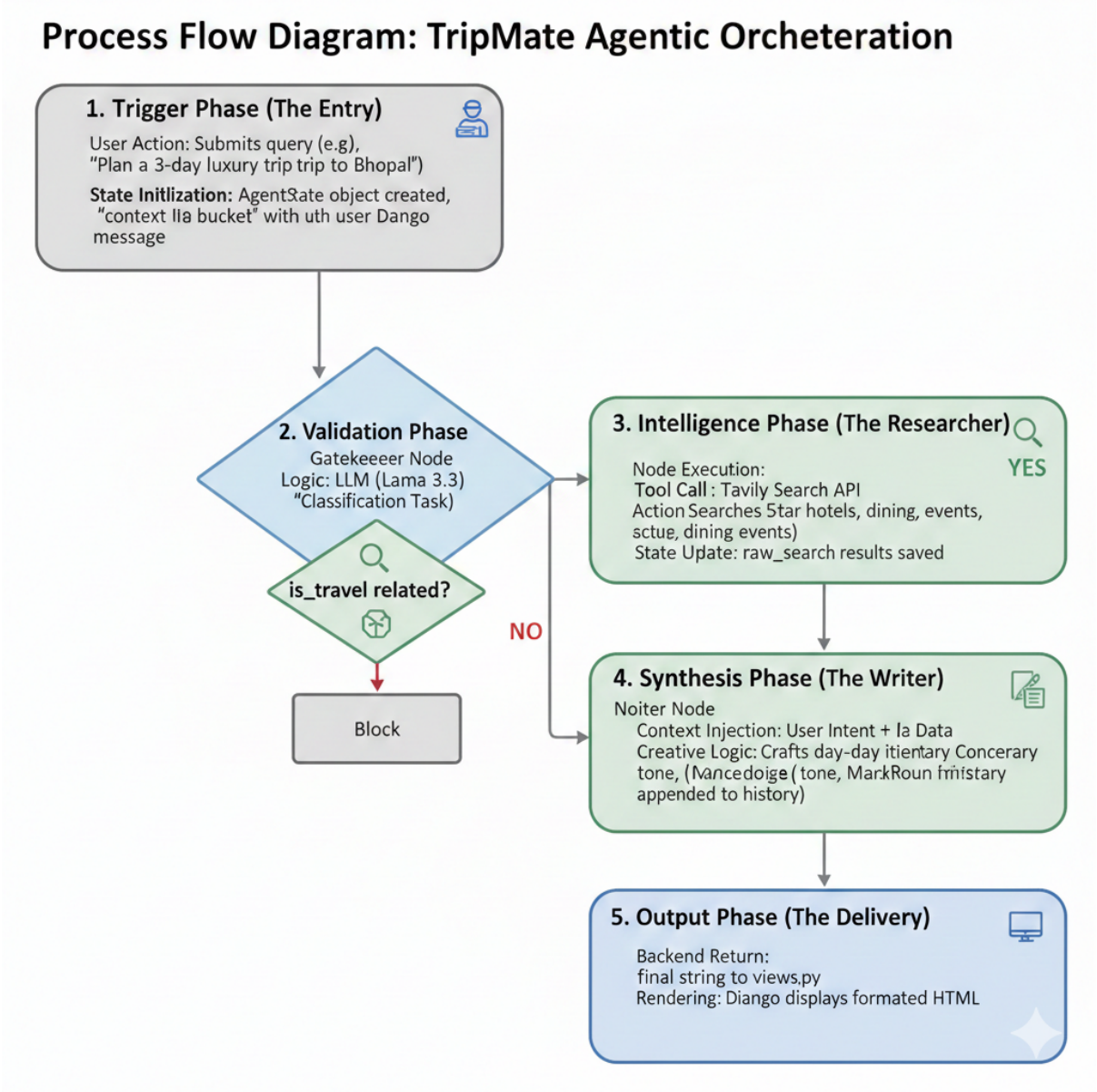
2.1. Key Features

- **Autonomous Research Agent:** Live web-browsing to find "Hidden Gems" and current events.
- **Domain Gatekeeping:** Automated rejection of off-topic or unsafe queries.
- **Dynamic Itinerary Synthesis:** Markdown-rendered schedules that adapt to user-specified durations.
- **Luxury-First Persona:** Tone and recommendation engine tuned for high-end hospitality.
- **Full-Stack Dashboard:** A responsive, elegant UI with real-time loading states.

2.2. Overall Architecture / Workflow

TripMate operates on a **Directed Acyclic Graph (DAG)** state machine. When a user submits a query, the state moves through the following nodes:

1. **Gatekeeper Node:** Analyzes the prompt to ensure it is travel-related.
2. **Researcher Node:** If validated, this node spawns a search process that queries the live web for the best hotels, restaurants, and activities.
3. **Writer Node:** Consolidates the research data and original user constraints into a professionally formatted Markdown itinerary.
4. **End Node:** Formats the final state and pushes it to the Django frontend.

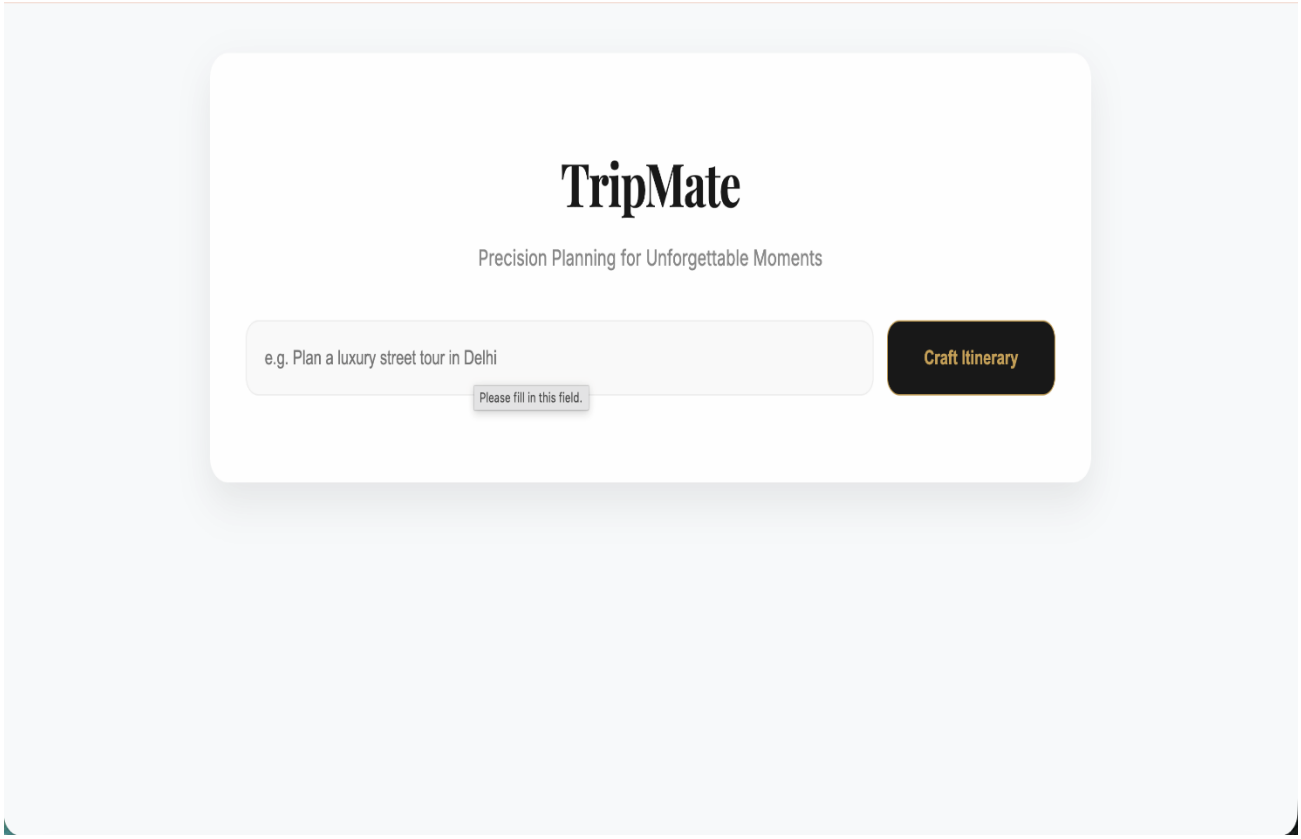


2.3. Tools & Technologies Used

- **Frameworks:** LangGraph (Orchestration), Django (Web Framework).
- **LLM & Inference:** Llama 3.3 70B via **Groq** (LPUs for ultra-fast processing)
- **APIs:** Tavily AI Search (RAG-optimized web search).
- **Frontend:** HTML5, CSS3 (Modern Luxury UI), Markdown-it (Rendering).
- **Environment:** Python 3.12, Virtual Environments (venv).

3. Results & Output

Screenshots / outputs



TripMate

Precision Planning for Unforgettable Moments

e.g. Plan a luxury street tour in Delhi

Craft Itinerary

2-Day Delhi Itinerary

Day 1: Exploring Historic Old Delhi and New Delhi

- 9:00 AM - 10:00 AM: Start the day with a hearty breakfast at a local restaurant, trying some traditional Indian cuisine like parathas or idlis.
- 10:00 AM - 12:00 PM: Join a **private Old and New Delhi tour** to get an in-depth understanding of the city's history and culture. This half-day tour will take you through the narrow streets of Old Delhi, showcasing its rich heritage and architectural marvels.
- 12:00 PM - 1:00 PM: Take a break for lunch at a renowned restaurant in Old Delhi, trying some local specialties like butter chicken or naan bread.
- 1:00 PM - 3:00 PM: Continue the tour, exploring the **Red Fort**, a UNESCO World Heritage Site and a symbol of India's rich history.
- 3:00 PM - 5:00 PM: Visit **Chandni Chowk**, one of the oldest and busiest markets in India, offering a sensory overload of sights, sounds, and smells.
- 6:00 PM - 8:00 PM: Enjoy a **food and heritage tour of Old Delhi**, sampling local street food and learning about the city's culinary traditions.

Day 2: Discovering New Delhi's Landmarks

- 9:00 AM - 10:00 AM: Start the day with a visit to the **India Gate**, a war memorial dedicated to the soldiers who lost their lives in World War I.
- 10:00 AM - 12:00 PM: Explore the **Rashtrapati Bhavan**, the official residence of the President of India, and learn about its history and architecture.
- 12:00 PM - 1:00 PM: Take a break for lunch at a nearby restaurant, trying some international cuisine or local delicacies.
- 1:00 PM - 3:00 PM: Visit the **Qutub Minar**, a UNESCO World Heritage Site and one of the tallest minarets in India, showcasing the city's rich cultural heritage.
- 3:00 PM - 5:00 PM: End the day with a visit to the **Lodhi Gardens**, a beautiful park with historical

The output is presented as a "Digital Itinerary" within a luxury-themed dashboard. Key visual elements include:

- **The Search Interface:** A minimalist gold-and-black search bar with an "Agentic Loader" showing the background reasoning process.
- **The Rendered Plan:** A clean, formatted itinerary with `<h3>` day headers, bulleted activity lists, and "Pro-Tips" sections.

3.2. Reports & Dashboards

The dashboard utilizes a **Responsive Grid Layout**. The results are containerized in a "Result Card" that uses the `|safe` filter to render HTML directly, ensuring that the AI's Markdown output is transformed into a readable web document.

3.3. Key Outcomes

- **Reduction in Research Time:** Users can generate a vetted 3-day plan in ~4 seconds compared to 2+ hours of manual searching.
- **Zero-Hallucination Guardrails:** By using RAG (Retrieval-Augmented Generation), the system successfully avoids recommending non-existent locations.
- **High Scalability:** The architecture allows for new agents (e.g., a "Budget Agent" or "Weather Agent") to be added to the graph without rewriting the core logic.

4. Conclusion

TripMate demonstrates the power of **Agentic AI** over traditional single-prompt models. By delegating tasks to specialized nodes and maintaining a persistent state, the system provides a level of precision and reliability required for the hospitality industry. Key learnings from this project include the importance of **Inference Speed** in agentic loops—where the choice of Groq was pivotal—and the necessity of **State Management** to ensure context is not lost between research and writing phases.

5. Future Scope & Enhancements

- **Multi-Modal Integration:** Allowing users to upload images of "vibe" or "style" (e.g., a picture of a specific resort) to guide the agent's research.
- **Real-Time Booking Integration:** Connecting to Amadeus or Skyscanner APIs to show live pricing and "Book Now" buttons.
- **Memory Persistence:** Implementing a database (like PostgreSQL) to allow the agents to remember a user's past preferences across multiple sessions.
- **Interactive Re-planning:** Adding a chat feature to the dashboard where a user can say "Make Day 2 less busy," allowing the agent to update the graph state dynamically.